

Glyph maps for spatio-temporal data in R

Seeing seasonality at every location with **sugarglider**, **cubble** and **GGally**

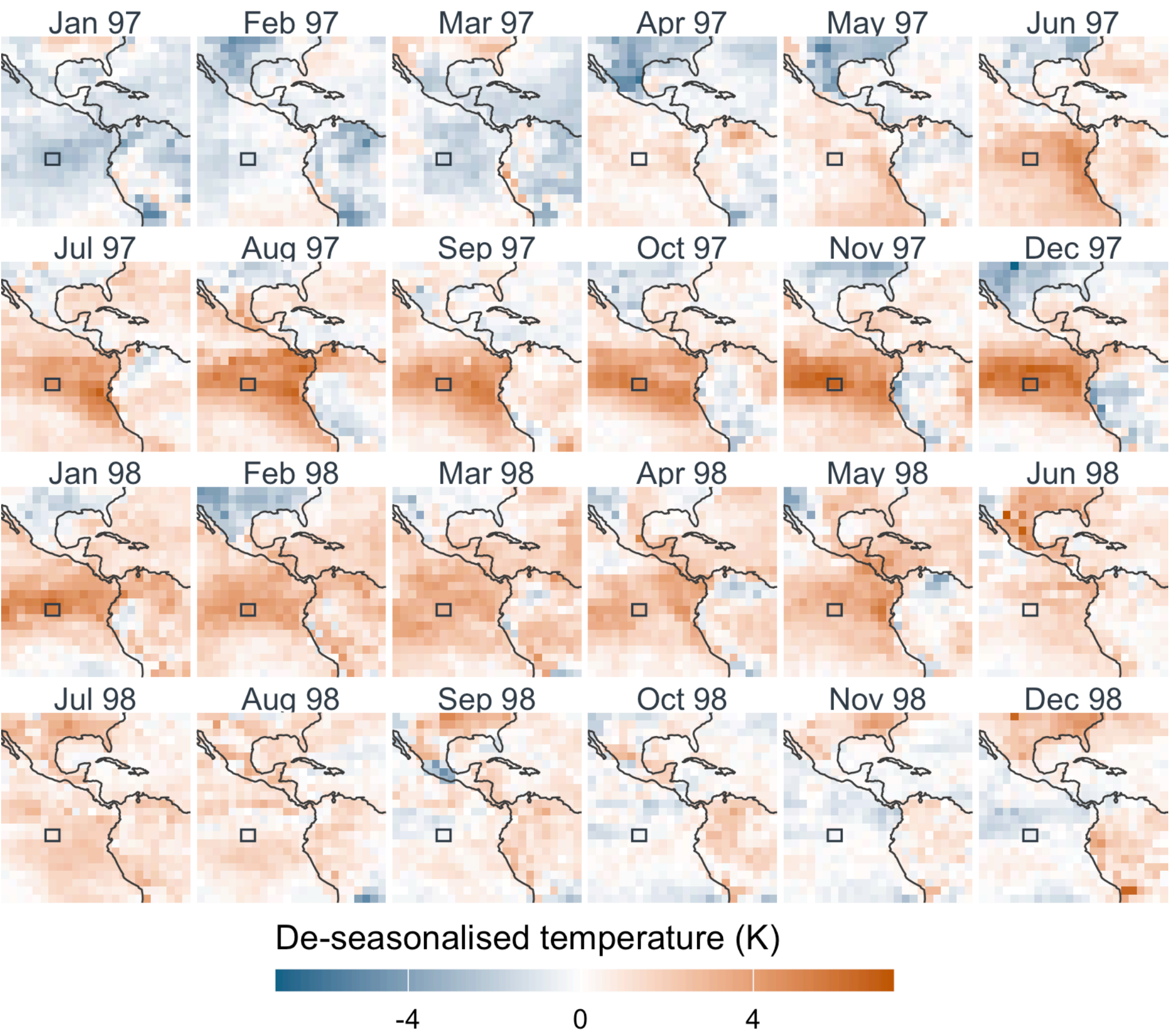
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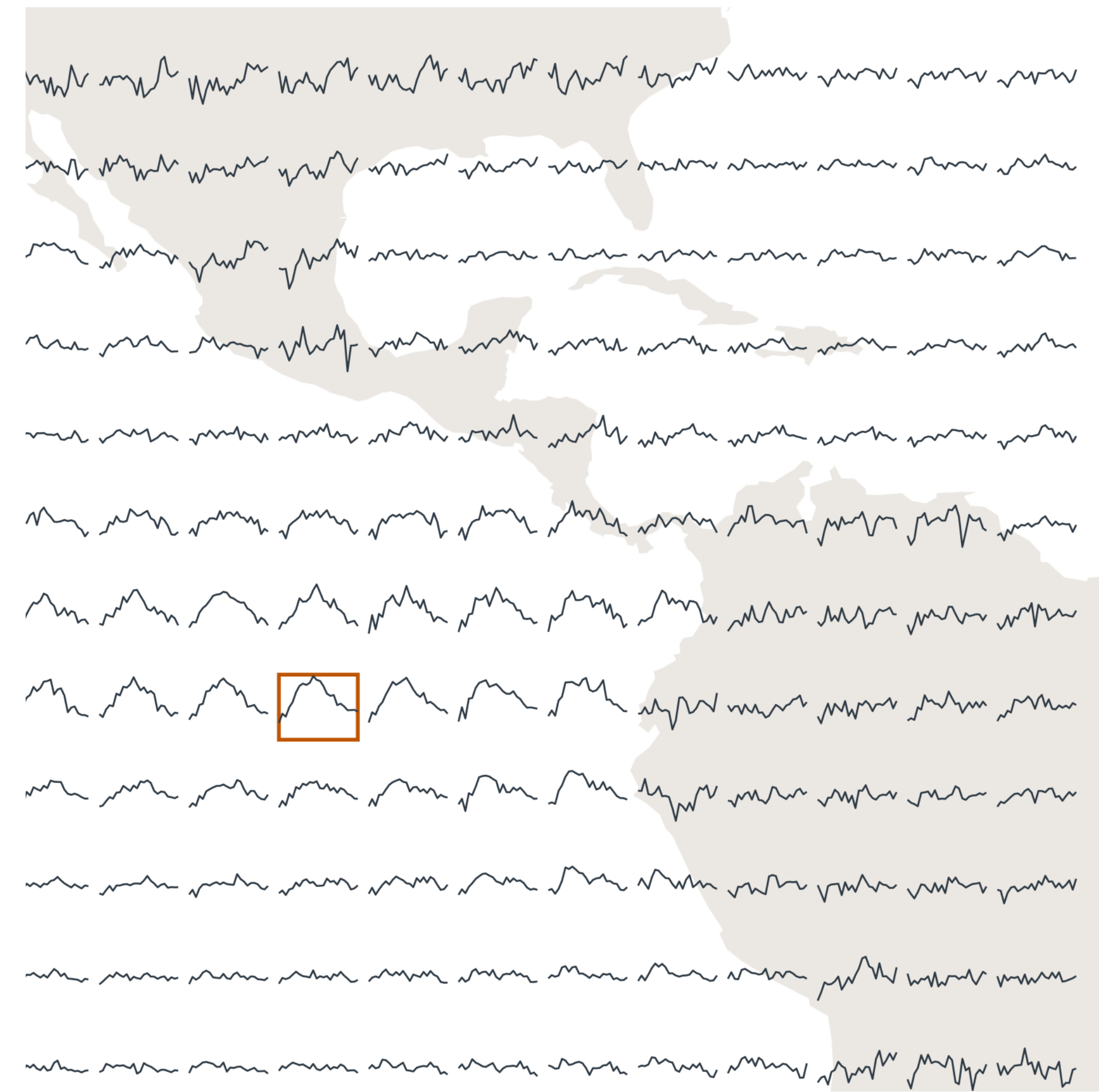
2012 From facets to a glyph map

Environmental data often come as **many time series at many locations**, commonly shown as maps faceted by time. Below, monthly de-seasonalised surface temperature over Central America for 1997–98.



Trace one location. Follow the boxed cell through the 24 panels to get its time series. A **glyph map** draws that series as a small plot at the cell's location (Wickham et al., 2012).

Do this for every location to get the full glyph map:



What the glyph map shows:

- Equatorial eastern Pacific glyphs share one hump peaking in late 1997 (the El Niño), fading away from the equator.
- Glyphs over land vary far more month to month than those over the sea.

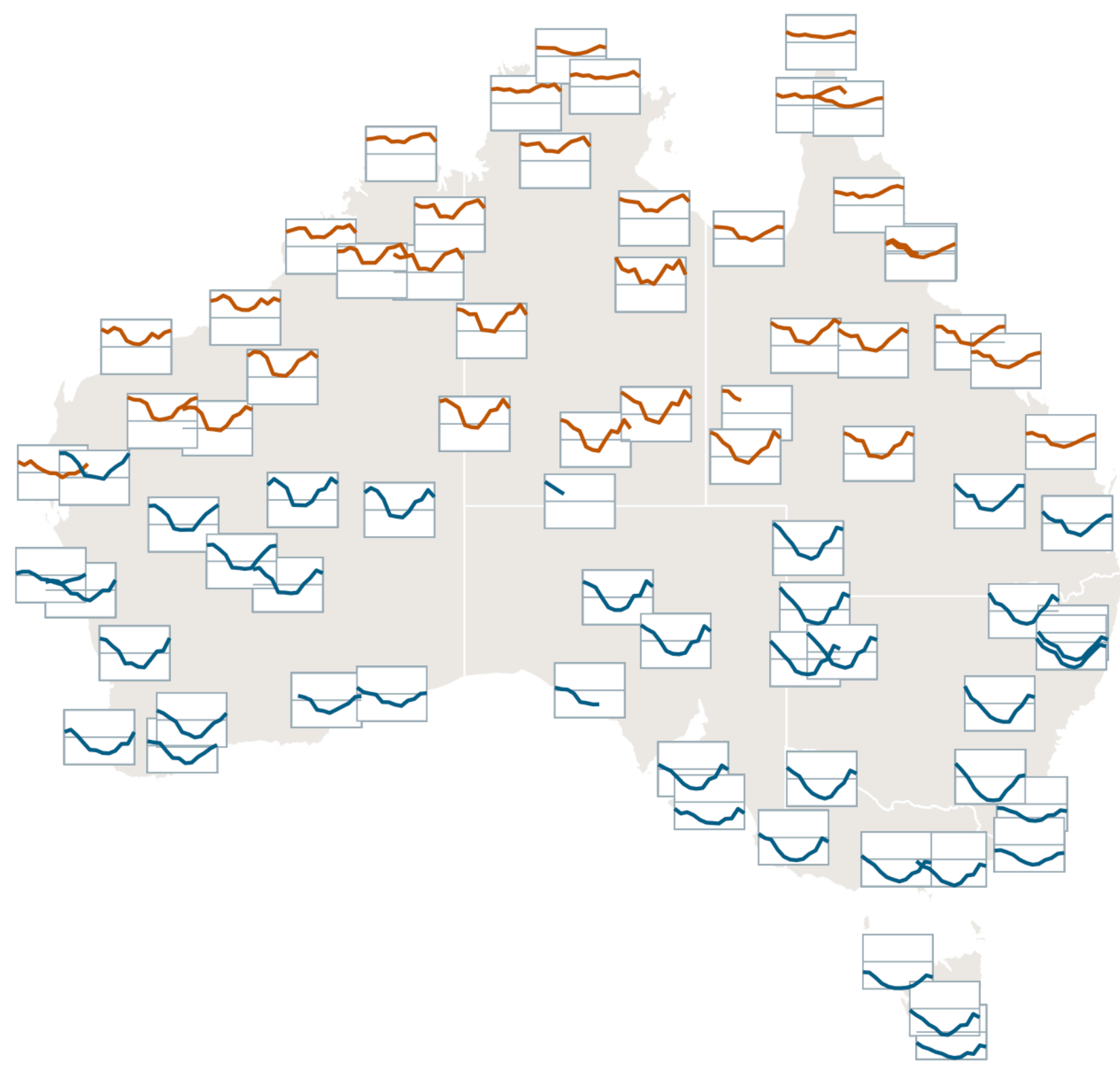
2023 Four new ggplot2 aesthetics

`cubble::geom_glyph()` moves the transformation inside a ggplot2 layer through **four new aesthetics**:

- `x_major`, `y_major`: where each glyph goes
- `x_minor`, `y_minor`: what each glyph draws

Because it is an ordinary layer, glyphs combine with base maps, colour and facets, and `geom_glyph_box()` / `geom_glyph_line()` add reference frames.

```
cb |>
  ggplot(aes(x_major = long, y_major = lat,
             x_minor = month, y_minor = tmax)) +
  geom_glyph_box() + geom_glyph()
```



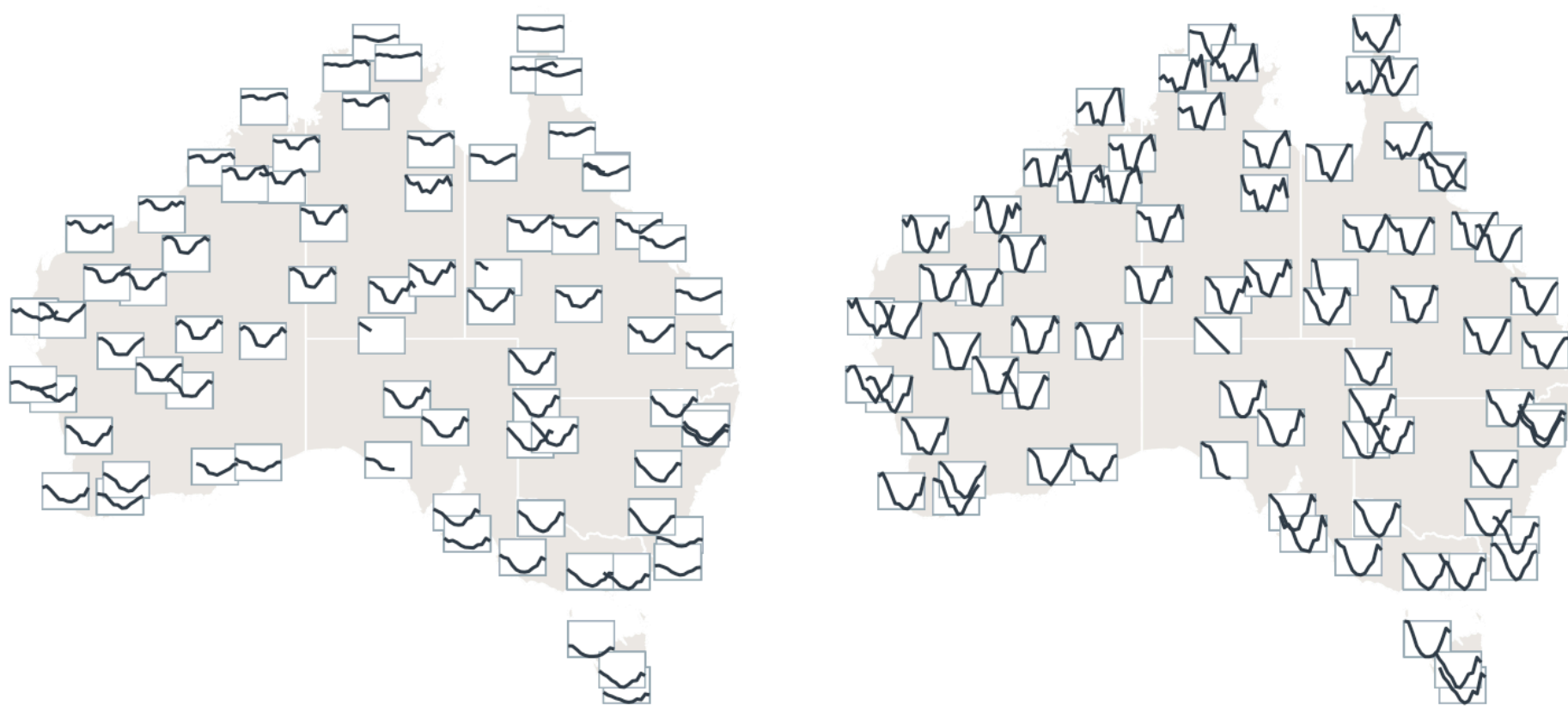
North of 25°S South of 25°S

Monthly average maximum temperature at about 90 stations. Southern stations swing strongly from summer to winter; northern stations stay hot all year.

The `global_rescale` argument sets what the glyphs compare: one shared scale for **magnitude**, or each glyph's own range for **shape**.

`global_rescale = TRUE`

`global_rescale = FALSE`



With a shared scale, northern glyphs look flat. With their own scale, every glyph fills its box and the small northern cycles become visible.

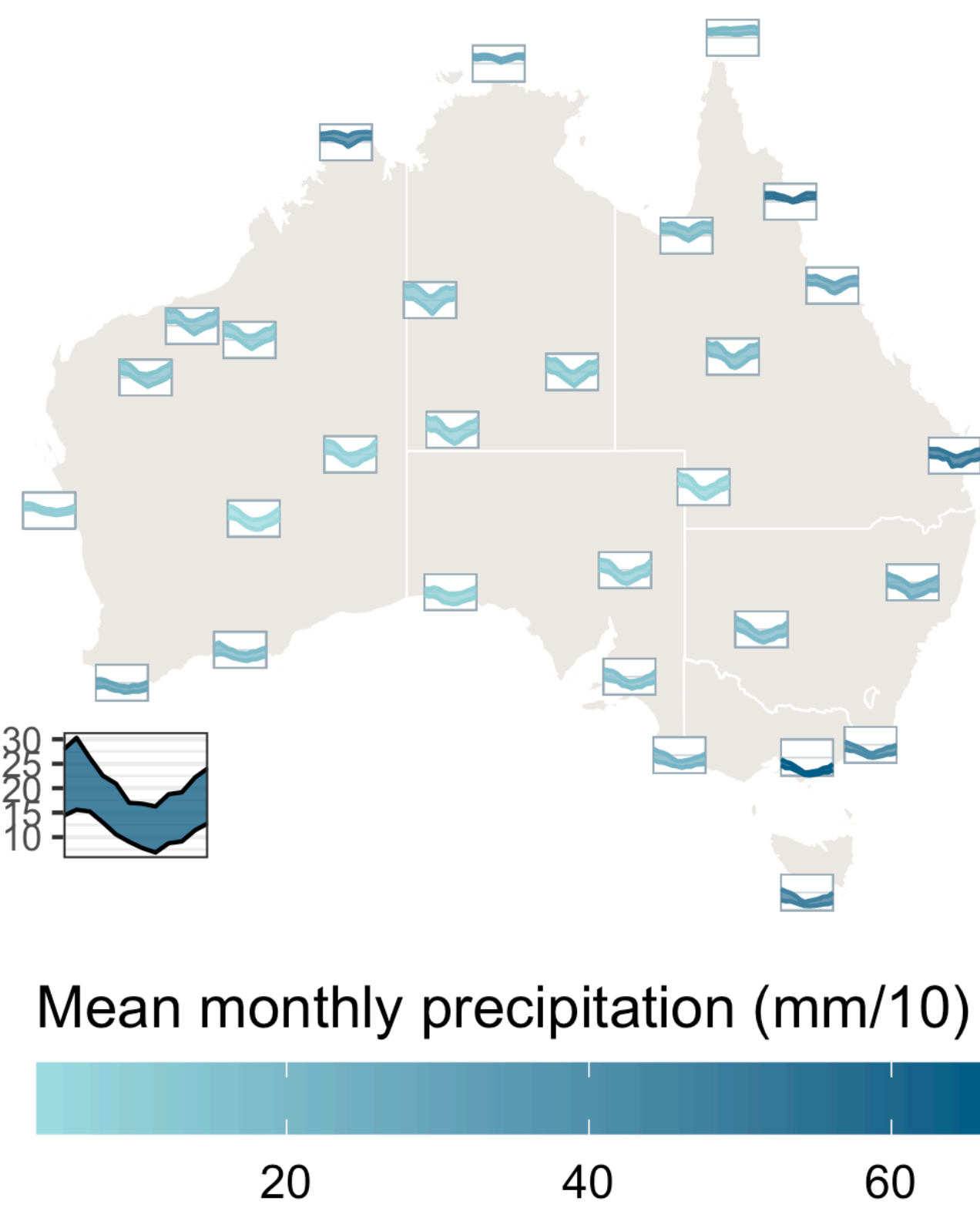
2026 Multivariate glyphs

A line glyph shows **one variable** over time, but many environmental summaries are **multivariate**: daily minimum *and* maximum, lower *and* upper quantiles, an estimate *and* its uncertainty. **sugarglider** adds aesthetics for a second minor variable, transformed the same way:

$$y_{\min} = y_{\text{major}} + \frac{h}{2} \tilde{y}_{\min, \text{minor}}$$
$$y_{\max} = y_{\text{major}} + \frac{h}{2} \tilde{y}_{\max, \text{minor}}$$

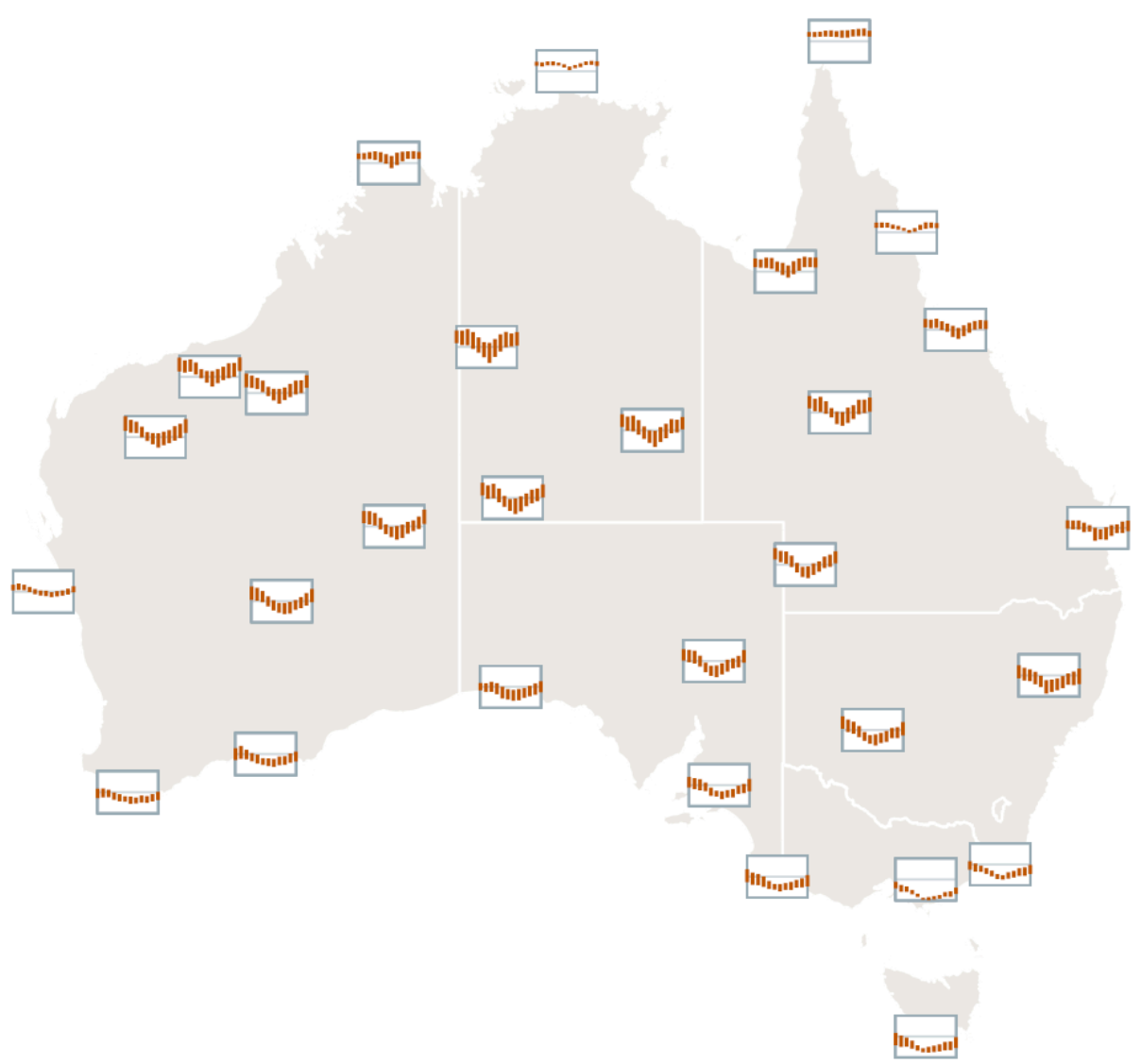
Ribbons fill the band between the two, here with a third variable, rainfall, as colour and an enlarged legend glyph to read the scale:

```
aus_temp |>
  ggplot(aes(x_major = long, y_major = lat,
             x_minor = month,
             ymin_minor = tmin,
             ymax_minor = tmax)) +
  add_glyph_boxes() + add_ref_lines() +
  add_glyph_legend() +
  geom_glyph_ribbon(aes(fill = prcp))
```



Segments draw one bar per time point from the same data:

```
aus_temp |>
  ggplot(aes(x_major = long, y_major = lat,
             x_minor = month,
             y_minor = tmin, yend_minor = tmax)) +
  add_glyph_boxes() + add_ref_lines() +
  geom_glyph_segment()
```



Wet tropical stations have narrow, flat bands; inland stations show wide daily ranges and deep winter dips.

Which one should I use?

	GGally	cubble	sugarglider
Interface	<code>glyphs()</code> → data frame	ggplot2 layer	ggplot2 layer
Glyph geometry	line	line	ribbon, segment
Reference box & line	✓	✓	✓
Global / local rescale	via <code>y_scale</code>	✓	✓
Polar glyphs	✓	✓	–
Glyph legend & theme	–	–	✓
Spatio-temporal data structure	–	✓	via cubble
Interactive (ggiraph, leaflet)	–	via crosstalk	✓

Take-home

- A glyph map shows each site's **temporal shape** and the **spatial pattern** in one plot.
- If your summary is a **range** (min–max, quantiles, uncertainty), use sugarglider's **ribbon** or **segment** glyphs.
- Use **global** rescaling to compare magnitude across sites and **local** rescaling to compare shape.
- All three packages work with **ggplot2**, so you can add base maps, colour and facets as usual.

Get started

```
install.packages(
  c("sugarglider", "cubble", "GGally"))
```

Paper



sugarglider



cubble



GGally



Wickham, Hofmann, Wickham & Cook (2012). Glyph-maps for visually exploring temporal patterns in climate data and models. *Environmetrics* 23(5), 382–393. [doi:10.1002/env.2152](https://doi.org/10.1002/env.2152)
Zhang, Cook, Laa, Langrené & Menéndez (2024). cubble: An R package for organizing and wrangling multivariate spatio-temporal data. *J. Stat. Softw.* 110(7). [doi:10.18637/jss.v110.i07](https://doi.org/10.18637/jss.v110.i07)
Po, Yang, Zhang & Cook (2026). sugarglider: Create glyph-maps of spatiotemporal data. *The R Journal* 18(3), 286–304. [doi:10.32614/RJ-2026-050](https://doi.org/10.32614/RJ-2026-050)